SIL-LAND: Segmentation Incremental Learning in Aerial Imagery via LAbel Number Distribution Consistency

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***Abstract*— Segmentation incremental learning (SIL) has received a lot of attention in recent years due to the ability to overcome the problem of catastrophic forgetting. Our study found that differences in label number distribution (LAND) affect the performance of SIL. Because the labels for pixels of the old category are marked as background when the model is trained on the new tasks, the LAND is inconsistent with static learning that is considered to be the upper bound on incremental learning, which hinders the mitigation of the catastrophic forgetting problem. In response to the above problems, we propose an incremental learning method named SIL-LAND, which improves the accuracy by making the LAND of our method close to that of static learning. From the perspective of high-level semantic labels, we propose the prototype update mechanism for the problem that nonadaptive representative prototypes ignore the sample diversity of semantic categories in remote sensing images. By compensating for the difference in LAND at the feature level, the distance between the prototype and the actual class center is reduced; aiming at the lack of semantic consistency between feature vectors and prototypes, we propose a similarity measure module to increase the intraclass similarity between the prototype and the corresponding feature vectors. From the perspective of one-hot labels, we propose label reconstruction, including foreground screening and background padding to make the number distribution of one-hot labels as close as possible to that of static learning. A series of experimental results demonstrates the effectiveness of our method.**

***Index Terms*— Incremental learning, label number distribution (LAND), label reconstruction (LR), prototype update, similarity measure.**

Manuscript received 25 April 2022; revised 12 July 2022; accepted 8 August 2022. Date of publication 17 August 2022; date of current version 2 September 2022. This work was supported by the National Natural Science Foundation of China under Grant 62171436. *(Corresponding author: Wenhui Diao.)*

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Digital Object Identifier 10.1109/TGRS.2022.3199584

I. Introduction

With the development of remote sensing technology, a large number of high-resolution remote sensing images [1] bring new inspiration and challenges to semantic segmentation. In general, the models proposed by most methods are optimized to use all data at once. When encountering new tasks/data, existing deep neural networks suffer from the problem of catastrophic forgetting [2], that is, after the model is trained on new data, the prediction results on the old tasks/categories drop significantly. To alleviate the problem of catastrophic forgetting, the easiest method is to retrain the model on a large amount of new and old data, which will lead to extreme stress in model training as the number of tasks increases. Incremental learning is an effective way to solve the above problems. As the new task/data comes, the model does not need to be retrained from scratch. Therefore, incremental learning has gradually received attention and has been widely studied.

The key to overcoming the problem of catastrophic forgetting is how to make the model better retain old knowledge when the model is fine-tuned on the new data [3], [4], [5], [6]. With regard to this, sparse and disentangled representation (SDR) [7] introduces prototypes to the segmentation incremental learning (SIL) for the first time. The high-dimensional feature vector named prototype is obtained by calculating the arithmetic average of the feature vectors for each semantic category, which approximately represents the high-level semantic labels. SDR achieves the preservation of old knowledge by preserving the same high-dimensional information between the new and old models. As the state-of-the-art method, SDR is selected as the baseline in this article. However, there are still feature gaps for different objects, scenes, sensors, and so on [8], [9] In particular, it is of significance to bridge the feature gap between incremental learning methods and static learning in remote sensing.

Our study found that the cause of catastrophic forgetting problem is related to the number distribution of labels, as shown in Fig. 1, that is, the number of labels in different categories used by all tasks, which is termed label number distribution (LAND) in this article. Subtle differences in LAND may lead to significant gaps in prediction results. In static learning, the model achieves an upper bound on performance because

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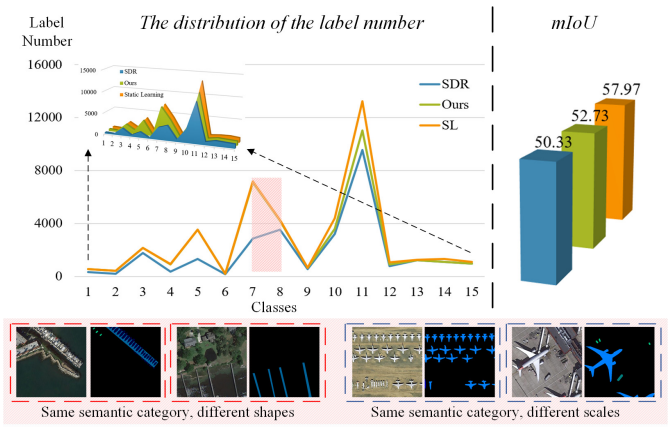


Fig. 1. Distribution of label number contained in different classes in the training dataset and the mIoU in the mib-ade-split protocol. The distribution of our method is closer to that of static learning. This means that the model is trained using almost all objects of different scales and shapes of the same semantic category in aerial imagery. Missing objects of any scale or shape may degrade the performance of this class. Best viewed in color.

the model is optimized using all the images and corresponding labels at once. Intuitively, the consistency of LAND between incremental learning methods and static learning is beneficial to alleviating the problem of catastrophic forgetting. However, in the typical incremental learning protocols, pixels whose labels are background may belong to certain classes specified in previous or later steps and vice versa. This causes the actual LAND to be different from the static learning. Specifically, the problem is manifested in three aspects in SDR.

A. Calculation of Prototypes Is Nonadaptive

In some remote sensing datasets, objects with the same semantic labels may have large differences in characteristics, resulting in fluctuations in the value of feature vectors. Also, the model parameters are gradually changed as the model is trained on new data, resulting in changes in feature vectors. Therefore, we argue that the prototype should be adaptively updated as new data arrives.

B. Feature Vectors Is Difficult to Maintain Semantic Consistency With Prototypes

Since the objects of the same semantic category in remote sensing images may vary significantly in shape and scale, the feature vectors of the same category are also different, which may cause feature confusion. Therefore, we argue that similarity constraints should be imposed between feature vectors and prototypes so that feature vectors of the same class and corresponding prototypes have a greater intraclass similarity.

C. Interclass Context in One-Hot Labels Is Missing

When the model is trained on the new data, the labels for pixels of old categories are marked as the background, resulting in the difference of LAND between SDR and static training, that is, the number of labels of the old category is reduced so that the relevant information of the images containing both the new categories and the old categories is

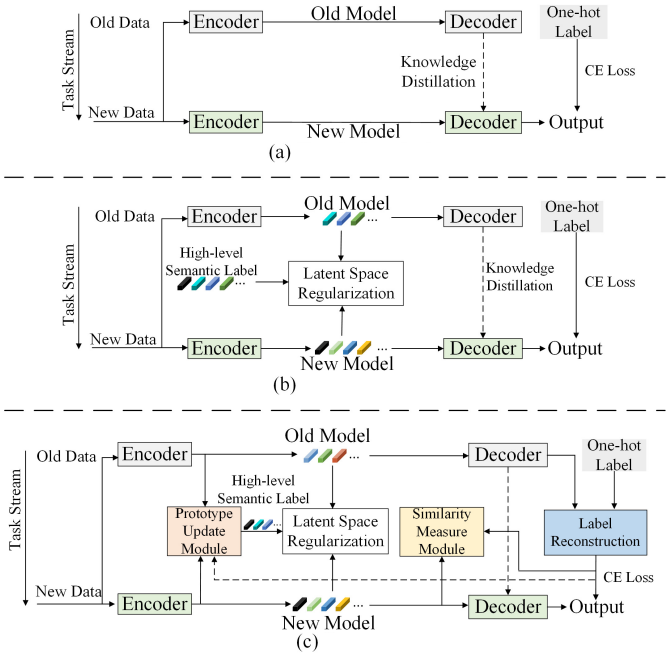


Fig. 2. Schematic of classical methods, baseline (SDR), and ours. (a) Classical methods pipeline. (b) Baseline pipeline. (c) Ours pipeline.

reduced, resulting in the interclass context in one-hot labels that is missing. Therefore, we argue that the relevant interclass context relationships in the labels should be preserved.

To overcome the above weaknesses, we propose an SIL framework for remote sensing images, denoted as SIL-LAND, as shown in Fig. 2(c). By introducing the semisupervised learning strategy to generate pseudo-labels, LAND is close to static learning. Specifically, for the first problem, in order to compute prototypes adaptively according to the old data and new data, we generate pseudo-labels corresponding to the unlabeled old-category data contained in the data on the current task. We propose the prototype update module that combines the feature vectors of the new model and the old model to compensate for the value shift of the class center calculated by the old model to realize the adaptive adjustment of the prototypes to the actual class center. For the second problem, the similarity measure module (SMM) is proposed to increase the similarity between feature vectors and the corresponding prototype. This module can realize that the intraclass similarity of prototype and corresponding feature vectors is greater than the interclass similarity. Last but not least, we propose label reconstruction (LR), including foreground screening and background padding to achieve joint learning with ground truth and pseudo-labels. We carry out comprehensive experiments on three challenging datasets and remarkable experiment results demonstrate the effectiveness of our approach.

Our main contributions in this article are the following.

1) We reveal and systematically study the important cause of catastrophic forgetting in incremental learning, that is, inconsistent LAND, and reveal three manifestations leading to this problem in classical methods. Based on

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the above findings, we proposed a novel incremental learning method named SIL-LAND.

2) Aiming at the problem of nonadaptive prototypes, we propose the prototype update module to make feature vectors of old categories in current data participate in prototype computation. By making the number distribution of high-level semantic labels involved in prototype computation close to that of static learning, the distance between prototypes computed from features vectors and class center is narrowed.
3) To address the problem of maintaining the semantic consistency between the feature vectors and the prototypes of the corresponding categories, we introduce an SMM to maintain the semantic consistency between the feature vectors and the prototypes. Feature vectors closer to the prototype ensure representation of the semantic categories.
4) For the issue of missing interclass context in one-hot labels, we employ the old model to make prediction for old categories to generate pseudo-labels. The model is trained end-to-end using the labels of new classes and the pseudo-labels of old classes in order to preserve the relationship of interclass context.

II. Related Works

In this section, we first review relevant works on incremental learning [10] that also referred as continual learning [11], sequential learning [12], or lifelong learning [13], [14]. Researchers mainly focus on the classification incremental learning in the early period and then gradually divert attention to detection and SIL. In addition, we also introduce some prototype-based methods.

A. Classification Incremental Learning

Incremental learning originated from the study of memory and forgetting mechanisms in cognitive neuroscience, dedicated to the study of models that continuously learn new knowledge without accessing old data. De Lange *et al.* [15], [16] grouped the classification incremental learning methods in three families: replay methods, regularization-based methods, and parameter isolation methods.

Replay methods [17], [18], [19], [20], [21] preserved information about old knowledge by preserving original samples or generating pseudo-samples. However, replay methods need to consider the burden of computing resources and measure the availability of old data. The regularization-based methods [3], [22], [23], [24], [25], [26], [27], [28], [29] affected the update of model parameters by adding new loss functions or changing the hyperparameters in the training so that the old knowledge is not covered by the new knowledge. Besides, parameter isolation methods [30], [31], [32], [33] also continue to develop as a branch of incremental learning methods, which is limited by the setting of task incremental. Our method belongs to regularization-based methods, which means that computational resources and parameters will not increase seriously.

B. Detection and SIL

With the development of incremental learning, researchers gradually divert attention from classification incremental learning to detection and SIL. Konstantin Shmelkov *et al.* [34] learned object detectors incrementally and proposed the loss function to balance the interaction between predictions of new and old classes. Xialei Liu *et al.* [35] analyzed cross-domain and cross-category continuous learning, breaking through the problem of models being trained and tested in the same domain. Dong *et al.* [36] proposed the use of unlabeled in-the-wild data to bridge the non-co-occurrence. Hao *et al.* [37] proposed a new dataset for class-incremental object detection. Zhou *et al.* [38] distilled selected anchors and proposals with the pseudo-positive-aware sampling strategy. There are still many methods [6], [39], [40] with outstanding contributions in the detection of incremental learning.

SIL methods are given as follows. RECALL [41] proposed two sources for replay data to recall knowledge of old tasks via GAN or Web Crawler. Incremental learning techniques (ILTs) [4] extended knowledge distillation to semantic segmentation, which is similar to LWF. UCD [42] and CAF [43] proposed new distillation losses for mitigating the problem of catastrophic forgetting. SSUL-M [44] introduces an additional unknown classes label assigned to the objects in the background, detected by the off-the-shelf saliency-map detector, and adopts the pseudo-labeling strategy. The pseudo-label-based approach [45], [46], [47] is a good way to overcome the semantic shift of the background class. ILSS [48] binarized the labels and then applies a binary cross-entropy loss for class-incremental learning. MiB [5] proposed the problem of semantic shift of background class in incremental learning for semantic segmentation for the first time. CLSI [49] presented two structure-preserving penalties to constrain the affinity matrices and align the high-frequency embeddings consistencies. Twin-auxiliary model [50] proposed diversity-distillation loss in order to retain the old knowledge of small objects and edges. SDR [7] first introduced prototyped in SIL and enforced the latent space consistent to alleviate the problem of catastrophic forgetting. In general, regularization-based methods occupy the mainstream in SIL methods. Our method takes SDR as the baseline because SDR has great potential to retain old knowledge not only through one-hot labels but also through prototypes.

C. Prototype-Based Methods

Prototypes have been widely used in various studies [51], [52], [53], [54], [55], [56], [57], [58], [59] and have been shown to represent key information for semantic categories. Concretely, ACFNet [52] proposed the concept of class center to extract global context from a class-level perspective. FeatMatch [53] was a semisupervised learning method that proposes complex transformations by using within-class and across-class prototypical representations. Gu *et al.* [54] proposed prototype-based approach for semisupervised classification. Prototype-based methods also play an important role in few-shot learning methods. PN [55] proposed a prototype network that can be used for few-shot learning. PANet [56]

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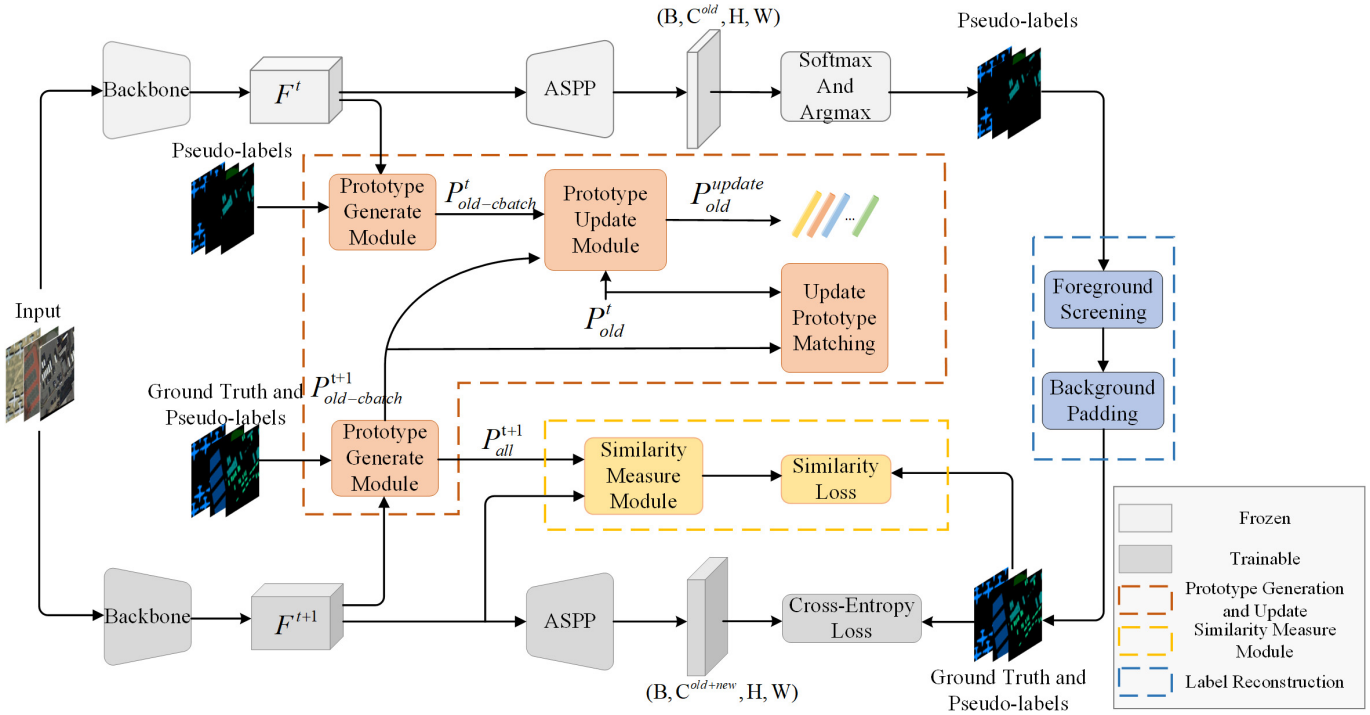


Fig. 3. Overall framework of our method. The new model and the old model are fed the data of the current task and its corresponding labels. The old model is frozen and the new model is trainable. Our method contains three contributions. Best viewed in color.

used the prototype as representative information and presented prototype alignment regularization in order to better utilize the information of the support set. SCNet [57] addressed the problem of incomplete construction of sample pairs with the foreground prototype only. PMMs [58] improved the prototype to represent the whole object so that each object is composed of multiple prototypes and each prototype contains a part of the object. Domain adaptation methods [60], [61] and feature representation methods [62] also utilized prototypes represent important information. As the first article introducing prototypes to SIL, SDR achieves a breakthrough in performance. Our method takes advantage of the prototypes' ability to represent the global information of the semantic class so that the knowledge of old data is not always forgotten when the model is optimized gradually.

III. Methodology

In this section, we describe our incremental learning method for semantic segmentation. First, we present the notation definition and the model structure. Second, we introduce the prototype generation and update (PGU) module in Section III-B. Then, SMM and LR are described in Sections III-C and III-D, respectively. Finally, we give some description on the training process.

A. Notation Definition and Network Architecture

Semantic segmentation can be defined as the pixel-level classification of images when all the training data are given at once. However, when data are gradually fed into the model in the form of the data stream, the model suffers from the problem of catastrophic forgetting. In recent years, SIL got a lot of attention on mitigating the catastrophic forgetting problem. Ideally, continual learning methods will not forget old knowledge without accessing old data. Our approach also focuses on SIL without using old data.

Formally, each incremental learning steps $T = (1, 2, \ldots, t, \ldots)$ is composed of the training set and the test data of a subset of dataset D. During the t step, we denote the training set as

$$D^t_{\text{train}} = \left(X^t_{\text{train}}, Y^t_{\text{train}}\right) \tag{1}$$

where $X^t_{\text{train}} = \{x_i\}^{K_t}_{i=1}$ denotes input images of size $H \times W \times 3$ and $Y^t_{\text{train}} = \{y_j\}^{K_t}_{i=1}$ is the label of size $H \times W$. K_t means the total number of images in the training set. Similarly, D^t_{test} denotes the test set at step t.

During the incremental step t, the classes currently learned by the model are denoted as $(0, 1, 2, \ldots, N_t)$. N_t denotes the number of foreground classes learned at step t. Labels for data that do not belong to the current task are marked as background c_{bg}. The labels contained in the training set at each phase are not coincident, that is, $Y^t_{\text{train}} \cap Y^s_{\text{train}} = c_{\text{bg}}$, where $t \neq s$ and t and s represent different steps. After all the data are learned in every step, the model makes predictions on all classes of the entire dataset without giving task-id, which is called class-incremental learning (CIL).

The model of our method is shown in Fig. 3, which consists of a backbone as an encoder and an atrous spatial pyramid pooling (ASPP) (with classifier) as the decoder. We define the model as $f^t_\theta = E^t \circ D^t$, where E means the encoder and D represents the decoder. Therefore, the feature map is denoted

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as $F^t = E^t(x) \in \mathbb{R}^{h\times w\times c}$ and the output is represented as $y_*^t = \text{argmax } f_\theta^t(x)$ of same size with the input images.

Specifically, at the t step, the model f_θ^t is trained to achieve $f_\theta^t : X_{\text{train}}^t \mapsto \mathbb{R}^{|H\times W\times N_t|}$ using the training set D_{train}^t, while the old data from D_{train}^1 to D_{train}^{t-1} are no longer available. The goal of incremental learning is to gradually learn new knowledge while retaining old knowledge under the premise that old data are not available, so as not to suffer from the problem of catastrophic forgetting.

B. PGU Module

Since the model is optimized causing parameters to change when encountering new data, and the objects of the same semantic category have different scales and shapes, we argue that adaptive prototypes are more suitable for the incremental learning paradigm. We investigate the prototype update mechanism in order to make the representative feature vector closer to the true class center. The diagram of prototype update module is shown in Fig. 4. Before describing the prototype update module in detail, we first introduce the prototype generation module.

1) Prototype Generation Module: Prototypes are generated in a variety of ways, where common methods include average the whole image [55] and mask average pooling (MAP) [57], [58], [63]. SIL focuses on the representative information extracted for each semantic class, so we use MAP to extract prototypes. Concretely, at each step t, the prototype of each class c can be defined as

$$p_c^t = \frac{1}{K^t}\sum_{K^t}\frac{\sum_u F_u^t \bigotimes \mathbb{I}\left(y_u^t = c\right)}{\sum_u \mathbb{I}\left(y_u^t = c\right)} \tag{2}$$

where F_u^t represents the feature vector of pixel u calculated from the forward propagation of the model f_θ^t. $\bigotimes$ is defined as elementwise multiplication and K^t means the number of all images in the training set during the step t. $\mathbb{I}(y_u^t = c)$ denotes an indicator factor. It equals to 0 when y_u^t belongs to the ith class and 0 in other case.

When the model is trained on the old data during step t, the prototypes of old classes are obtained. We denote the prototypes that have learned as P_{old}^t during the step t, which is given as follows:

$$P_{\text{old}}^t = \left[p_1^t \,\|\, p_2^t \,\|, \ldots, \| \, p_{N_1+N_2+\cdots+N_{t-1}}^t\right] \tag{3}$$

where $\|$ represents the concatenation operation. When the new data arrive, the old model is frozen and the new model is trainable. The prototypes of all classes that have been learned so far are represented as P_{all}^{t+1} during the step $t+1$

$$P_{\text{all}}^{t+1} = \left[p_0^{t+1} \,\|\, p_1^{t+1} \,\|\, p_2^{t+1} \,\|, \ldots, \| \, p_{N_1+N_2+\cdots+N_t}^{t+1}\right]. \tag{4}$$

Prototypes provide representative information for all classes learned by each task. In SDR, the prototypes of the current task are calculated and saved according to the ground truth, without considering past and future knowledge. However, this approach does not consider the high probability case that new data contain old knowledge. Thus, we propose an update mechanism for prototypes by exploiting pseudo-labels widely used in semisupervised learning.

2) Prototype Update Module: Objects of the same semantic category in remote sensing datasets have various morphologies and scales. Each instance is important when some classes contain a few images. Missing any of the images may cause performance degradation. Furthermore, as new data come, the model is updated in order to achieve good performance on both new and old data simultaneously. Therefore, the prototype should be updated adaptively as new data arrive so that the prototype is close to the true class center. On the premise of fitting the LAND of the static learning as much as possible, the prototype update mechanism proposed by us performs the update iteration of the class center according to the new data.

The distance between the prototype calculated from the old data and the real class center is called semantic shift [64]. Assume that the knowledge of class c is first learned by the model during step s, and the model is currently learning the knowledge of step t. The semantic shift at this time can be defined as

$$\Delta_c^{s\to t} = p_c^t - p_c^s. \tag{5}$$

The reason for the semantic shift is the knowledge contained in the new data after step t and the fine-tuning (FT) of the parameters of the model from step s to step t. Inspired by semisupervised learning, the model predicts pseudo-labels for the pixels that are labeled as background. To compensate for the semantic shift, we propose a prototype update mechanism. First, considering that model f_θ^s has the best predictive ability for old knowledge, the feature map obtained by feeding new data into model f_θ^s should be considered. Then, due to the FT of parameters caused by the learning of new knowledge by model f_θ^{s+1}, the knowledge of old tasks contained in model f_θ^{s+1} should also serve as a reference. Therefore, the feature maps output by f_θ^s and f_θ^{s+1} should participate in the update of the prototype to obtain semantic offset compensation at the same time. For the input data of the current batch, the semantic offset compensation is given as follows:

$$p_c^{\text{batch}} = \frac{1}{K^{s+1}}\sum_{K^{s+1}}\frac{\sum_u F_u^s \bigotimes \mathbb{I}\left(y_*^s = c\right)}{\sum_u \mathbb{I}\left(y_*^s = c\right)} \oplus \frac{1}{K^{s+1}}\sum_{K^{s+1}}\frac{\sum_u F_u^{s+1} \bigotimes \mathbb{I}\left(y_*^{s+1} = c\right)}{\sum_u \mathbb{I}\left(y_*^{s+1} = c\right)} \tag{6}$$

where p_c^{batch} represents the semantic offset of category c from the data of the current batch and $\oplus$ represents the arithmetic mean operation.

It is important to note that we consider situations where pseudo-labels may cause misclassification. Although pseudo-labels contain the most information related to the knowledge of old tasks currently available, it is inevitable that the prediction results of pseudo-labels have a small amount of error with the ground truth, which means that a certain category determined by the pseudo-labels is different from the ground truth. When the distance between feature vectors and the previously calculated prototype is quite different, we think that it is probably due to the false prediction of the pseudo-labels. Therefore, we propose to set the hyperparameter ρ in the prototype update mechanism and decide whether to update

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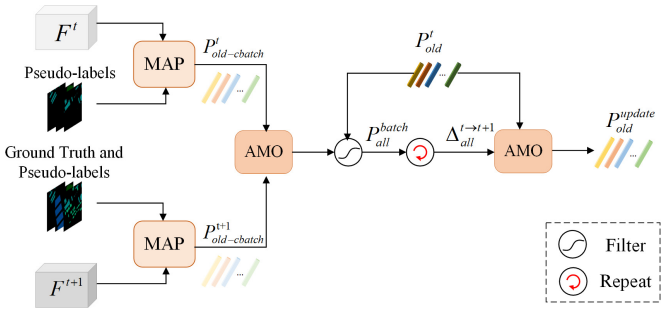


Fig. 4. Diagram of prototype update module. MAP and AMO denote mask average pooling and arithmetic mean operation, respectively. When the distance is shorter than ρ, the filter is turned on; otherwise, the filter is turned off. Repeat means that the aforementioned process is repeated as the model is trained. The statistical average of $P^{\text{batch}}_{\text{all}}$ obtained by each batch is expressed as $\Delta^{t\to t+1}_{\text{all}}$ that means semantic offsets for the entire training process.

each step of the model by comparing the distance between the feature vectors and the previously calculated corresponding prototype.

Specifically, we get the prototype p^{batch}_c for each category based on the current batch of input images. When the sum of the distances between p^{batch}_c and the prototype p^t_c calculated by the previous task is less than the hyperparameter ρ, p^{batch}_c participates in the update of the prototypes

$$p^{\text{batch}}_c = \begin{cases} p^{\text{batch}}_c, & \text{if } \sum \left(p^{\text{batch}}_c - p^t_c\right) < \rho \\ \text{None}, & \text{if } \sum \left(p^{\text{batch}}_c - p^t_c\right) \geq \rho \end{cases} \tag{7}$$

where None means that the prototype of category c will not be updated in the current batch.

According to (7), p^{batch}_c is determined whether participates in the update of the prototype of this category and all the prototypes calculate the statistical average with P^t_{old} to get $P^{\text{batch}}_{\text{all}}$. Then, repeat the above process and calculate $P^{\text{batch}}_{\text{all}}$ for each batch, which we call repeat in Fig. 4. All the prototypes of old classes are represented as $P^{t+1}_{\text{old_cbatch}}$ and $P^t_{\text{old_cbatch}}$. We denote $\Delta^{t\to t+1}_c$ as semantic offset of class c after the model is optimized in all epochs. After an epoch, the output of the prototype update module is denoted as $P^{\text{update}}_{\text{old}}$.

Compensation for the prototype can be achieved by calculating the semantic shift in the training process of the model at each phase, namely

$$p^{s\to t}_c = p^s_c \oplus \Delta^{s\to s+1}_c \oplus \Delta^{s+1\to s+2}_c \oplus \cdots \oplus \Delta^{t-1\to t}_c \tag{8}$$

where $p^{s\to t}_c$ represents the updated prototype of a semantic category after the model is trained during the step t. The updated prototypes also consider the effects of old knowledge contained in new data and FT of model parameters and theoretically have a better ability to represent semantic category information.

C. Similarity Measure Module

Prototypes are the average of feature vectors with representative information of semantic categories. The same representative information in the new model helps the model retain the knowledge of the old task, which is beneficial to alleviating the catastrophic forgetting problem. Since objects of the same

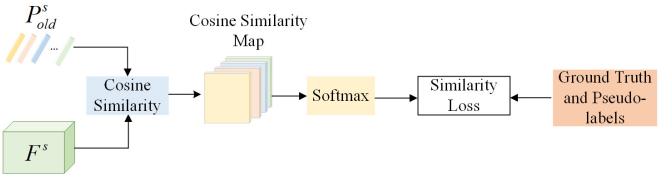


Fig. 5. Illustration of SMM.

semantic category in remote sensing images have different shapes and scales, feature confusion may occur. In order to keep the semantic consistency between the feature vectors and the prototype of the same semantic category, we propose an SMM so that the feature vector of the same category and the class center of its corresponding category should have a greater similarity. The diagram of the SMM is shown in Fig. 5. The purpose of the SMM is to achieve the transition from Fig. 6(a) to Fig. 6(b). The feature vectors and the prototype of its corresponding class should have a greater similarity. The similarity between class centers and feature vectors that are not of the same class should be as small as possible. Inspired by computing the cosine similarity of features and prototypes in few-shot learning methods [56], [65], [66], we calculate the cosine similarity between the feature map $F^t \in \mathbb{R}^{h\times w\times C}$ and the prototypes $p_c \in \mathbb{R}^{C\times 1}$

$$S_c = \cos(F^t, \varepsilon(p_c)) \tag{9}$$

where $S_c \in \mathbb{R}^{h\times w}$ denotes the similarity map between p_c and every feature vector of feature map and $\varepsilon(p_c) \in \mathbb{R}^{h\times w\times c}$ means the broadcast of p_c. $\cos(\cdot)$ represents the cosine similarity function, which can be defined as

$$\cos\left(f^t_i, p_c\right) = \frac{f^t_i p_c}{\left\|f^t_i\right\|_2 \cdot \|p_c\|_2} \tag{10}$$

where $f^t_i \in \mathbb{R}^{C\times 1}, i = 1, 2, \ldots, hw$, means the feature vector. The larger the cosine similarity, the closer the semantic consistency between the feature vector and the corresponding prototype. Then, the similarity map S is calculated by the softmax function. $S \in \mathbb{R}^{h\times w\times C}$ represents the similarity map calculated by cosine similarity between the feature map and the prototype of each category. We upsample the normalized similarity map S to $H \times W \times C$ and calculate the similarity loss

$$\mathcal{L}_{\cos} = -\frac{1}{|x|}\sum_{u\in x} \log\left(\text{Upsample}\left(\text{softmax}\left(S^t_x(u, y_u)\right)\right)\right) \tag{11}$$

where Upsample represents the upsampling operation through bilinear interpolation. The SMM constrains the relationship between the prototype and the feature vectors so that the cosine similarity between the feature vector and the prototype of the corresponding category is greater. For classes with similar semantic categories, a prototype will be more discriminative.

D. Label Reconstruction

SIL aims to continuously learn new knowledge while maintaining old knowledge without accessing any old data, which has multiple protocol settings. For example, in the

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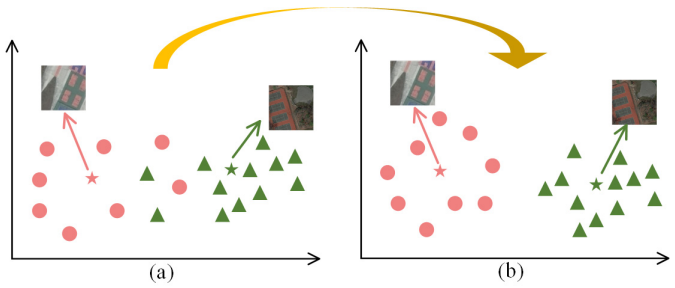


Fig. 6. Illustration of similarity between the prototype and feature vectors after dimensional reduction. Pink and green represent two different categories. The circles represent feature vectors. The five-pointed star indicates the prototype corresponding to the semantic category. An SMM aims to realize the transformation of relationship between the prototype and the feature vectors from (a) to (b), which means that the similarity of feature vectors and corresponding prototype is greater. (a) Prototype and feature vectors of two classes without SMM after training in the task1. (b) Prototype and feature vectors of two classes with SMM after training in the task1.

disjoint [5], [7] protocol, the RGB images contain the categories of the old tasks that have been seen before and the current task. The label of the new classes is foreground category, while the label of the old classes learned by the old task and background class is background. In addition, overlap [5], [7], [34] is also a commonly used incremental learning protocol. Overlap setup refers to each training step containing all images of current classes, but only the pixel positions are labeled as the foreground. Images in the current step may contain instances of old and future classes. In short, regardless of the protocol settings, the pixels whose label belongs to the category of old tasks are assigned as background on the current task. It is reasonable to assign the label corresponding to the old class as the background, as this avoids repeated labeling of the same class.

However, the labels of old categories are marked as background, which leads to the difference of LAND with static learning, that is, the relevant information of the images containing both the new and the old categories in the training set is reduced, resulting in the lack of context between classes. Inspired by semisupervised learning methods, we aim to recover the interclass context in labels.

Pseudo-label-based methods can make simple and effective label predictions on unlabeled pixels and have been extensively validated in semisupervised learning. We propose LR to change the labels of old data labeled as background in new data. The model is trained end-to-end using the labels of the new classes and the pseudo-labels of the old classes to preserve the interclass context of related semantic classes while avoiding the classifier's bias toward the new class.

Pseudo-labels are derived from the prediction results of the model trained in the previous step. Specifically, LR consists of two steps: foreground screening and background padding. The foreground labels in the new data are artificially labeled, which is theoretically more accurate. Thus, in the foreground screening process, we use the foreground labels to filter the false predictions of the pseudo-labels. Pseudo-labels are derived from the prediction results of the model trained in the previous step. Specifically, LR consists of two steps:

Algorithm 1 Training Procedure of the Proposed Method

Input:
$T = (1, 2, \ldots, t, \ldots)$: The step of CIL;
D^t_{train}: The training set during the step t;
$p_c^{s\to t-1}$: The prototypes computed by previous tasks;
Output:
θ_t: The continual learning model;
$p_c^{s\to t}$: The prototypes computed by previous tasks and current task;

if t=1 **then**
  Random Initialize θ_t
  Forward propagation using train set D^1_{train}
  Get the total loss $\mathcal{L}_{ce} + \mathcal{L}_{cos}$ (Eq. 13 + Eq. 11)
  for $c \in y^t_*$ **do**
    Compute the prototypes p^1_c according to labels Y^1_{train};
  end for
  Backward propagation and update the parameters θ_1
else
  Initial θ_t from θ_{t-1}
  Forward propagation of model θ_{t-1} using training set D^t_{train} in order to obtain pseudo-labels
  Forward propagation of model θ_t using train set D^t_{train} to obtain the feature map $F^t = E^t(D^t_{train})$
  for $c \in y^t_*$ **do**
    Compute the prototypes p^t_c of current task according to labels Y^t_{train}
  end for
  Get the similarity loss $\mathcal{L}_{cos}$ (Eq. 11)
  Get the loss of $\hat{\mathcal{L}}_{ce}$ (Eq. 13)
  for $c \in y^1_* \cup y^2_* \cup \cdots \cup y^{t-1}_*$ **do**
    Get the semantic shift $\Delta_c^{t-1\to t}$
    Compute the prototypes $p_c^{s\to t}$ of previous tasks according to $\Delta_c^{t-1\to t}$
  end for
  Get the loss of $\hat{\mathcal{L}}_{SDR}$
  Get the loss of $\mathcal{L}_{tot}$ (Eq. 15)
  Backward propagation and update the parameters θ_t
end if
return θ_t and $p_c^{s\to t}$

foreground screening and background padding. Because the foreground labels in the new data are artificially labeled, which is theoretically more accurate, thus in the foreground screening process, we use the foreground labels to filter the false predictions of the pseudo-labels. Then, in the background padding, we replace the pixel labels labeled as background in the train set with the pseudo-labels predicted by the old model

$$\hat{y}^t_u = \begin{cases} y^t_u, & \text{if } y^t_u \neq c_{\text{bg}} \\ y^{t-1}_*, & \text{if } y^t_u = c_{\text{bg}} \text{ and } y^{t-1}_* \neq c_{\text{bg}} \\ c_{\text{bg}}, & \text{if } y^t_u = c_{\text{bg}} \text{ and } y^{t-1}_* = c_{\text{bg}} \end{cases} \tag{12}$$

where $\hat{y}^t_u$ denotes the new labels that combine the pseudo-labels and correct labels of pixel u in images x. SDR applies L_{ce} to learn new knowledge. According to the ground truth and pseudo-labels, L_{ce} is changed in our proposed

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SIL method, which not only plays the role of learning new knowledge but also retains old knowledge, which can be defined as

$$\hat{\mathcal{L}}_{\text{ce}}(\theta_t) = \frac{1}{N_t} \sum_{(x,y)\in D^t} l_{\text{ce}}(\theta_t; x, \hat{y}) \tag{13}$$

where $\hat{y} = y$ during the first step and l_{ce} is the cross-entropy loss, which can be defined as

$$l_{\text{ce}}(\theta_t; x, y) = -\frac{1}{|x|} \sum_{u\in x} y_u \cdot \log q_x^t(u, y_u) \tag{14}$$

where y_u represents the one-hot ground truth of pixel u in image x and $q_x^t(u, y_u) = f_{\theta_t}(x)[u, c]$.

Ideally, there is no need to manually relabel a large amount of data for the old category pixels in the new data. From the perspective of LAND, we propose that LR includes foreground screening and background padding so that the model can simultaneously learn the knowledge of new and old categories to preserve the interclass context relationship. Under the premise of satisfying the ideal conditions of continuous learning, the model is not exposed to any obfuscated knowledge, which helps to alleviate the problem of catastrophic forgetting.

E. Integrated Optimization

From the perspective that inconsistent LAND affects the mitigation of the catastrophic forgetting problem, our method contains three contributions. The first contribution is an adaptive prototype update method. The second contribution is to constrain the correlation between feature vectors and prototypes through a point-level similarity measure. Finally, we preserve interclass context through LR. The total loss function of our method is given as follows:

$$\mathcal{L}_{\text{tot}} = \lambda_{\text{ce}}\hat{\mathcal{L}}_{\text{ce}} + \lambda_{\text{SDR}}\hat{\mathcal{L}}_{\text{SDR}} + \lambda_{\text{cos}}\mathcal{L}_{\text{cos}} \tag{15}$$

where $\hat{\mathcal{L}}_{\text{SDR}}$ refers to the loss function after changing the prototype p_c calculated in the previous task in the prototype matching loss to our improved prototype $p_c^{s\to t}$. The updated prototype matching loss is expressed as update prototype matching in Fig. 3. The training process of our method is shown in Algorithm 1.

IV. Experiments

In this section, we first introduce the overview of datasets. Then, we present implementation details, including training details and evaluation metrics. Next, the compared methods are briefly described. Finally, we analyze the experimental results, including ablation experiments, quantitative analyses, and qualitative analyses.

A. Datasets

We carry experiments on the iSAID [67] dataset, the GaoFen Challenge dataset [68], and the Pascal VOC2012 [69] dataset. The iSAID dataset is a large-scale dataset, for instance, segmentation and semantic segmentation in aerial images. The iSAID dataset contains 655 451 object instances for 15 important and commonly occurring categories across 2806 high-resolution images. In the experiments, we use the published 1411 images to participate in training and test on 458 images of the validation set. The rest of the 937 image annotations in the test set are unavailable. The distinctive characteristics of iSAID are the foreground–background imbalance and long tail of instance count. The GaoFen Challenge dataset is the dataset of track4 in the 2020 GaoFen Challenge, which is provided by the Gaofen-2 satellite with 0.8-m resolution. There are nine categories, including road, building, shrub and tree, lawn, land, water body, vehicle, impervious ground, and others. We split the training and test sets with a ratio of 4:1. The training set contains 759 RGB images, and the test set contains 189 RGB images with the size of 2000 × 2000. We crop the image to 512 × 512 patch for the iSAID dataset and GaoFen challenge dataset. Pascal VOC2012 contains 10 582 and 1449 images in the training set and test set, respectively. Pascal VOC2012 has 21 different classes, including background. The code is available at https://github.com/ljx97/SIL-LAND.

B. Implementation details

1) Training Details: As in SDR [7], our model applies Deeplab-v3+ [70] with ResNet-101 [71] pretrained on the ImageNet [72] as our backbone. The initial learning rate is set to $7e^{-3}$ for the iSAID dataset and GaoFen Challenge dataset and $1e^{-3}$ for the VOC dataset during the first step, and the learning rate is set to $7e^{-4}$ for the iSAID dataset and GaoFen challenge dataset and $1e^{-4}$ for VOC dataset in subsequent tasks, which is the same as other incremental learning methods [7]. We use the polynomial decay with power 0.9 to update the learning rate for each iteration. The model is optimized with stochastic gradient descent (SGD), momentum 0.9, and weight decay $1e^{-4}$. Following [7] and [34], we crop all images to 512 × 512 for all datasets and apply the same data augmentation with them. To ensure fair comparisons, the hyperparameter λ in our method is the same as SDR. λ_{cos} and λ_{ce} are equal to 1. All the experiments are carried out on a 16-GB NVIDIA Tesla P100 GPU for 30 epochs with a batch size of 8. All the compared methods are the same in training sources, experiment settings, and a number of epochs.

2) Evaluation Metrics: For measuring the segmentation performance, we adopt mean intersection over union (mIoU) as an evaluation metric. In incremental learning, the mIoU denotes the average of IoU over all classes that the model has learned. The IoU is defined as

$$\text{mIoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{16}$$

where TP, TN, FP, and FN are the number of true positives, true negatives, false positives, and false negatives, respectively.

C. Compared Methods

To measure the effectiveness of our method, we compare our method with some classical and state-of-the-art methods. Since our method is a regularization-based method and does not retain any original samples or generate samples by GAN, the compared methods are also some regularization-based methods, which are given as follows:

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1) Offline: It, also called static learning, refers to the model that is optimized to use all images and their corresponding labels, which is not an incremental learning method. Offline is seen as the upper bound of the performance of incremental learning methods.

2) Fine-Tuning: It is one of the classic incremental learning methods. The idea of FT comes from transfer learning. FT refers to the fact that the model is trained on new data and labels without any constraints on the knowledge of the old tasks. Obviously, models that are not constrained to retain the knowledge of old tasks will suffer from catastrophic forgetting problems. The model performs well on the categories of new task, but the knowledge of the old tasks is almost completely forgotten.

3) LWF [3]: Following [5], [7], [41], and [73], we use LWF as compared method. LWF is one of the regularization-based methods, which introduces the idea of knowledge distillation into classification incremental learning for the first time. LWF applies a distillation loss to constrain the model so that the distributions of the soft labels between the teacher model and the student model are close.

4) LWF-MC: It is the single-headed version of LWF, which is always used together with LWF as a compared method [5], [7], [41], [73].

5) ILT [4]: It is an SIL method based on knowledge distillation, which designs a couple of novel distillation losses. ILT does not store any samples of previous datasets. The distillation strategy we adopt in the experiments is consistent with SDR. ILT alleviates the problem of catastrophic forgetting.

6) CIL [74]: It optimizes the distillation loss function to achieve incremental learning for segmentation tasks without requiring images or labels of old task.

7) MiB [5]: It proposes the distribution shift problem of background class encountered in incremental learning for semantic segmentation tasks and improves the distillation loss to solve it. MiB can perform well on both old and new tasks without resorting to any samples of old tasks.

8) SDR [7]: It introduced prototypes as representative feature vectors into SIL for the first time and significantly outperforms state-of-the-art methods. SDR also does not need to retain any samples of old tasks and belongs to one of the regularization-based methods. Our experiments use SDR as the baseline method, and the hyperparameters used are consistent with it.

D. Ablation Study

In order to explain how our method works, we conduct a series of ablation experiments on th iSAID dataset with 10-5 task.

1) Ablation Study for Each Component: The experiment results for each component are shown in Table I. The model can achieve 50.33% mIoU when trained with the SDR method. When adding the prototype update mechanism, the mIoU increased by 0.9%. This proves that the update mechanism of prototypes in incremental learning is useful and necessary. Since the amount of data in incremental learning shows an increasing trend, our method considers the diversity of samples and update the prototype in order to make the prototype more representative of the semantic information of one class. Furthermore, when applying PGU and SMM or PGU and LR simultaneously, our method has 1.34% and 1.77% mIoU improvement over the baseline method, respectively. Finally, it yields the best 52.73% mIoU cooperated with the three components. The surpassing performance of different combinations of the three contributions demonstrates the effectiveness of our proposed method for SIL.

TABLE I
ABLATION STUDY FOR EACH COMPONENT

Baseline	PGU	SMM	LR	mIoU(%)
✓				50.33
✓	✓			51.23
✓	✓	✓		51.67
✓	✓		✓	52.10
✓	✓	✓	✓	**52.73**

2) Ablation Study for Prototypes Update: The results on different prototypes update mechanisms are shown in Table II. The hyperparameter ρ is equal to 500, 1000, and 1500 and not limited to the threshold. Specifically, the first row indicates that the feature vectors corresponding to the pseudo-label predicted output by the teacher model. Compared with the baseline method of 50.33% mIoU, this prototype update mechanism has a better performance under all four hyperparameters. Similarly, the new network is constantly learning about the knowledge of old tasks from the old network, so the second row shows the experiment results of reordering the feature vectors using the pseudo-labels predicted by the new network. Compared with the baseline method, it also has good performance. Based on this, we consider using the feature vectors corresponding to the predicted results of both the old model and the new model. Since the predictions of both the old network and the new network are correct in most of the regions of the segmentation, using the feature vectors of both helps to reduce the proportion of incorrect predictions by the model. The experimental results are shown in the last row. The mIoU in this case peaked at a hyperparameter ρ of 500.

In addition, we design to reduce false predictions by setting a threshold τ on pseudo-labels or sampling the predictions of pseudo-labels, which are shown in the third row and fourth row. The prototype update mechanism under the pseudo-label sampling condition is unstable. The prototype update mechanism works only when the hyperparameter is greater than 1000 and the prototype update reduces the mIoU of the baseline in other cases. Moreover, when the predicted probability of the pseudo-label by the teacher model is greater than 0.9, the feature vectors of the corresponding position participate in the prototype update, which performs well too. To sum up, the prototype update mechanism is necessary for the incremental learning process due to the diversity of sample and FT of

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TABLE II
ABLATION STUDY FOR PROTOTYPES UPDATE MODULE

Method	None	500	1000	1500	Max
teacher_feature	50.93	50.81	50.82	50.59	50.93
student_feature	50.94	50.88	50.44	50.75	50.94
sample	50.15	49.79	50.92	49.99	50.92
thr0.9	50.17	50.46	51.00	50.66	51.00
teaandstu_feature	50.81	51.23	50.73	50.87	**51.23**

TABLE III
ABLATION STUDY FOR SMM

Method	None	500	1000	1500	Max
Base + PGU + LR	51.16	51.40	51.97	51.30	51.97
+ $\mathcal{L}_{cos}$_2softmax	49.48	51.82	50.05	50.99	51.82
+ $\mathcal{L}_{cos}$_12softmax	**52.32**	**52.73**	**51.84**	**52.36**	**52.73**
+ $\mathcal{L}_{cos}$_2sigmoid	49.67	50.56	50.23	50.12	50.56
+ $\mathcal{L}_{cos}$_12sigmoid	48.69	48.18	47.40	48.86	48.86

model parameters. Also, we achieved excellent performance by using all the feature vectors filtered from the prediction results of the old model and the new model to participate in the prototype update.

3) Ablation Study for SMM : When calculating the similarity map, we experiment with softmax or sigmoid operation on the similarity probability map. The experimental results are shown in Table III. The first row shows the results without using the SMM. The best result is 51.97% mIoU. In the second and third rows, the similarity map uses the softmax operation to normalize the similarity probability. In the fourth and fifth lines, the similarity map uses the sigmoid operation to process the similarity probability. The second and fourth rows indicate that the similarity loss is used only in new tasks. The third and fifth lines indicate that the similarity loss is calculated at each step. Normalizing the similarity probabilities using the softmax operation is significantly better than normalizing using the sigmoid operation. Since the cross-entropy loss is calculated between the similarity map and the label, the closer the similarity is to 1, the smaller the loss. However, sigmoid does not have this constraint. As shown in Table III, we can see that adding similarity loss to each step achieves the best results. As shown in Table IV, the cosine similarity between feature vectors and prototypes of all classes is greater than the baseline. The distinguished results proved that it is more reasonable to apply similarity constraints to the prototypes and feature vectors of all tasks. To sum up, applying the SMM in each step helps to update the prototype so that the semantic similarity between the feature vector and the corresponding prototype is higher.

TABLE IV
AVERAGE COSINE SIMILARITY OF ALL CLASSES ON THE iSAID VALIDATION SET

Method	old	new	all
SDR	70.0370	70.9960	70.3567
Ours	70.6580	71.3200	70.8787

TABLE V
ABLATION EXPERIMENT FOR BACKGROUND PROTOTYPE

Method	mIoU (%)		
	old	new	all
Bkgupdate	52.75	51.78	52.45
Ours	**53.02**	**52.08**	**52.73**

4) Ablation Study for Background Prototype: The experiment results are shown in Table V. It is observed that the mIoU of all classes of bkgupdate (Bkgupdate) reaches 52.45%, which is lower 0.28% than that of the model that only updates prototypes of foreground classes (Ours). The mIoU of old classes and new classes achieves 52.75% and 51.78%, respectively, which is both lower than ours. Intuitively, SIL encounters the problem of semantic shift of background class, and the positions of pixels belonging to the background classes are changed. Thus, the prototype of the background class contains complex information. When the prototype of the background class is updated during training, a larger proportion of the feature vectors involved in the calculation of the prototype of the background class will come from the foreground class, causing the prototype of the background class to be unstable. It can be noticed that the bkgupdate method is still 2.12% mIoU (50.33% $\rightarrow$ 52.45%) higher than the baseline method, even if it has a negative effect on the final result. This result proves the effectiveness of our method in avoiding catastrophic forgetting.

5) Ablation Study for Class Orders: To demonstrate the effect of different class orders on the class-incremental learning method, we conduct experiments based on the iSAID dataset. Specifically, we randomly generate four orders for the 15 foreground categories contained in the iSAID dataset in the 10-5-mib-split protocol. The experimental results are shown in Table VII. The result of order 4 is the best among the four orders, achieving mIOU $=$ 55.08%, which is 1.10% mIoU higher than the SDR. In the experimental results of order3, our method outperforms the baseline method by more than 3% mIoU. In other orders, our method also outperforms the baseline method by about 2% mIoU. This proves that our incremental learning method is effective in retaining old knowledge while learning new knowledge regardless of the order of categories.

6) Ablation Study for the Methods of Calculating Similarity: The experiment results are shown in Table VI. We conduct

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TABLE VI
ABLATION EXPERIMENT FOR CALCULATING SIMILARITY

Order	Dist		mIoU			Δ
	Euclid	Cosine	old	new	all	
order1	✓		52.67	51.53	52.31	
		✓	53.02	52.08	52.73	+ 0.41
order2	✓		53.39	41.74	49.75	
		✓	52.82	44.09	50.09	+ 0.34
order3	✓		49.53	28.64	43.00	
		✓	51.40	27.69	43.99	+ 0.99
order4	✓		54.16	39.19	54.16	
		✓	61.18	41.67	55.08	+ 0.92
order5	✓		57.86	42.32	53.00	
		✓	57.94	42.85	53.22	+ 0.22

experiments on five orders with Euclidean distance (Euclid) and cosine distance (Cosine) based on our proposed methods. The five orders are the same as in Table VII. It is observed that our proposed methods with cosine distance have better performance, which is higher 0.5% mIoU than our method with Euclidean distance. The largest gap caused by different similarity measurement methods comes from order3, reaching 0.99% mIoU. Moreover, no matter which methods are used to calculate the similarity, the results are better than the baseline method. It can be concluded that when applying SMM to constrain the relationship between feature vectors and prototypes, semantic consistency can be maintained better.

7) Ablation Study for Hyperparameter: The experimental results are shown in Fig. 7. The baseline method appears as a black dashed line. After adding the prototype update mechanism, the experimental results are shown in the green solid line. When the hyperparameter is greater than 200, mIoU achieves an overall rise, reaching a high level when the threshold is 500 and 900. When the hyperparameter is less than 100, the update of the prototype only considers the feature vectors that are similar to the previous prototype, and the prototype has almost no offset, so the value floats up and down the baseline. Our method is shown by the red solid line, which achieves the best performance when the hyperparameter is 500, which exceeds the baseline by more than 2% mIoU. When the hyperparameter is in the range of 800–1000, the experiment results are unstable, which may be because the feature vector with a far distance from the prototype causing incorrectly affects the update of the prototype. There are two reasons for the distant similarity between prototype and feature vector. On the one hand, it may be due to the misclassification of pseudo-labels. On the other hand, it may be due to the FT of network parameters to make the network more suitable for new tasks. However, in this case, our method still outperforms the baseline method by more than 1% mIoU. To sum up, the hyperparameters between 400 and 700 can achieve prominent

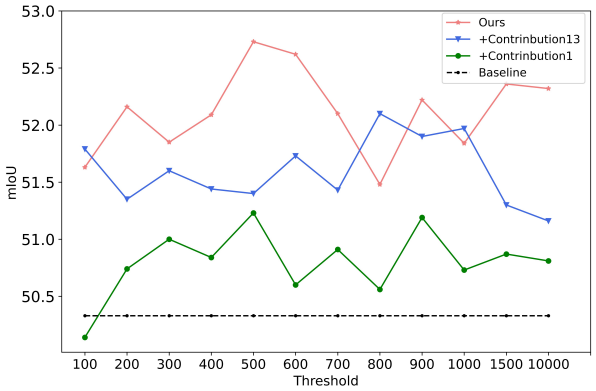


Fig. 7. Ablation study for hyperparameter ρ with thresholds between 100 and 10 000.

experimental results, which not only enriches the feature representation of the prototype but also avoids the feature vector that is far similar to the prototype from incorrectly guiding the update of the prototype.

E. Quantitative Analysis

1) Experiments on the iSAID Dataset: We designed three experimental settings: 10-5, 15-1, and 5-5-5. 10-5 means we train the model on the ten classes for the first task and add the remaining five classes in the next task. There are four training protocols: disjoint, overlap, random, and mib-ade-split. MiB proposed disjoint and mib-ade-split. Disjoint refers to the fact that the images for the current task do not contain future classes, and the labels only contain new classes. Mib-ade-split assigns data to each task in proportion to the proportion of the image that contains the new class. Overlap was proposed in [34], which means that the training set in the current task consists of all images containing the new class. The labels of the classes in the new task are marked as foreground, and the labels of other classes are background. Random refers to randomly dividing a dataset into disjoint parts. During each step, only the labels of the new class are labeled as foreground, and the labels of other classes are labeled as background. In an ideal incremental learning approach, data are fed into the model as a stream of data, and there is no way to estimate whether the data contain previous or future images. Random is a more realistic experimental protocol that does not require restrictions on the categories contained in the image.

First, the experimental results for the 10-5 task setting are shown in Table VIII. Category order is in alphabetical order. Offline refers to the fact that the model is trained on all data at once, which is generally regarded as an upper bound on the performance of incremental learning. As expected, FT performs well on new task and is at the forefront of almost all methods. However, the performance of FT on the old task is almost the worst. This is due to the fact that FT fine-tunes the parameters for the data of the new task and does not adopt any method to retain the old knowledge, so it encounters a serious problem of catastrophic forgetting. CIL and FT had similar effects. When adding the knowledge distillation strategy, LWF performs better on the old task, and the catastrophic forgetting

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TABLE VII
ABLATION FOR CLASS ORDERS. BEST IN BOLD

Method	IoU Per Category (%)																mIoU (%)		
	BG	**BD**	**BC**	**BR**	**GTF**	**HA**	**HC**	**LV**	**PL**	**RD**	**SH**	**SV**	**SFB**	**ST**	**SP**	**TC**	**old**	**new**	**all**
SDR	98.08	66.32	44.19	28.01	43.80	38.90	30.23	45.65	76.06	39.75	55.82	33.04	28.42	58.74	37.97	80.29	51.53	47.69	50.33
Ours	98.57	55.17	37.02	32.01	45.46	42.33	31.10	52.21	77.51	53.41	58.43	32.61	50.24	57.29	41.97	78.28	**53.02**	**52.08**	**52.73**
	BG	**BD**	**BR**	**HA**	**LV**	**RD**	**SV**	**SFB**	**ST**	**SP**	**TC**	**BC**	**GTF**	**HC**	**PL**	**SH**			
SDR	97.99	67.68	27.41	30.53	47.96	41.84	35.15	32.74	64.17	13.71	84.84	48.19	25.68	18.39	73.69	55.03	49.46	**44.20**	47.81
Ours	98.22	68.09	27.32	35.32	51.58	35.76	37.13	34.26	64.63	43.44	85.25	45.64	31.84	15.14	73.89	53.93	**52.82**	44.09	**50.09**
	BG	**HC**	**LV**	**PL**	**RD**	**SH**	**SV**	**SFB**	**ST**	**SP**	**TC**	**BD**	**BC**	**BR**	**GTF**	**HA**			
SDR	98.00	0.00	41.71	31.35	14.02	57.47	35.31	61.63	61.15	34.23	84.50	29.17	27.36	19.19	20.51	38.95	47.22	27.04	40.91
Ours	98.05	0.08	48.96	38.73	29.89	59.07	37.91	65.22	62.37	40.35	84.76	20.80	37.15	19.84	21.02	39.65	**51.40**	**27.69**	**43.99**
	BG	**BD**	**BR**	**GTF**	**HA**	**HC**	**PL**	**SH**	**SFB**	**ST**	**TC**	**BC**	**LV**	**RD**	**SV**	**SP**			
SDR	98.74	69.51	28.60	53.24	38.44	31.51	76.26	57.20	64.66	64.54	83.82	40.25	50.62	32.49	33.17	40.65	60.59	39.44	53.98
Ours	98.79	64.01	29.13	55.11	41.96	29.41	77.12	58.67	68.66	65.50	84.57	42.14	50.69	42.27	33.84	39.42	**61.18**	**41.67**	**55.08**
	BG	**BD**	**BC**	**GTF**	**HB**	**HC**	**PL**	**RD**	**SH**	**SP**	**TC**	**BR**	**LV**	**SV**	**SFB**	**ST**			
SDR	98.34	68.54	52.34	48.13	37.95	14.97	75.55	24.91	57.11	44.63	83.86	21.76	50.25	33.26	35.57	59.53	55.12	40.07	50.42
Ours	98.58	67.03	52.79	44.13	41.96	25.36	77.04	36.25	59.49	49.33	85.36	22.56	50.61	32.85	47.20	61.04	**57.94**	**42.85**	**53.22**

TABLE VIII
PERFORMANCE COMPARISONS ON THE iSAID DATASET FOR 10-5 CLASS-INCREMENTAL LEARNING SCENARIOS. BEST IN BOLD

	mib-ade-split			disjoint			overlap		
Method	**old**	**new**	**all**	**old**	**new**	**all**	**old**	**new**	**all**
FT	9.04	48.52	21.38	9.01	**52.83**	22.70	8.87	53.92	22.95
LwF	50.53	47.63	49.62	48.46	47.74	48.23	55.09	52.43	54.26
LwF-MC	19.03	29.62	22.34	17.06	29.41	20.92	17.69	31.25	21.93
ILT	50.43	45.59	48.92	47.97	44.58	46.91	52.06	44.67	49.75
CIL	8.88	41.42	19.05	8.98	45.31	20.33	8.86	47.86	21.05
MiB	47.52	48.59	47.85	46.90	50.17	47.92	52.54	**54.71**	53.22
SDR	51.53	47.69	50.33	48.44	49.41	48.74	54.98	51.66	53.95
Ours	**53.02**	**52.08**	**52.73**	**50.17**	49.28	**49.89**	**56.43**	51.57	**54.91**
Offline	56.91	60.30	57.97	56.91	60.30	57.97	56.91	60.30	57.97

problem is improved. Because LWF-MC is a single-head prediction, the performance is significantly lower than that of LWF. ILT and MiB also perform well on both new and old tasks. In particular, MiB achieves the optimal performance on new tasks in the overlap scenario. SDR is the suboptimal method among all methods, only worse than our method. From the view of overall performance, our method significantly outperforms all compared methods in all scenarios and is the closest to performance of offline. Although the performance of our method on the new task is not satisfactory, only mib-ade-split surpasses other baseline methods, but the performance of our method on the old task has reached the best and surpassed the baseline method by about 2% mIoU in all scenarios. In addition, each task contains all images of the new category in the overlap scenario, so the overall performance is better than the other two scenarios.

Then, we conduct experiments under the 15-1 task setting in three scenarios, including random, disjoint, and overlap. Likewise, offline represents a nonincremental learning method and is considered an upper bound on the performance of incrmental learning. In terms of old task performance, FT, LWF-MC, and CIL are the worst among all methods in random scenarios. FT transfers knowledge from old tasks to new tasks, and only focusing on the knowledge of new tasks leads to catastrophic forgetting. LWF-MC is slightly better than FT, and some categories can still be distinguished. CIL and FT perform almost identically on the old task. LWF, ILT, and MIB perform well on the old tasks, reaching 48.88%, 45.61%, and 48.90% mIoU, respectively. Almost every category has a large improvement in IoU. The performance of the baseline method on the old task is only worse than our method, reaching 49.56% mIoU. Our method achieves the best performance on

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TABLE IX

PERFORMANCE COMPARISONS ON THE ISAID DATASET FOR 15-1 CLASS-INCREMENTAL LEARNING SCENARIOS. BEST IN BOLD. OFFLINE IS NOT AN INCREMENTAL LEARNING METHOD, WHICH IS GENERALLY CONSIDERED TO BE AN UPPER BOUND ON PERFORMANCE IN CLASS-INCREMENTAL LEARNING

Protocols	Method	IoU Per Category(%)																mIoU (%)		
		BG	BD	BC	BR	GTF	HA	HC	LV	PL	RD	SH	SV	SBF	ST	SP	TC	old	new	all
Random	FT	97.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	73.46	6.49	73.46	10.67
	LwF	98.52	53.23	38.92	30.11	50.26	35.34	20.23	50.72	72.21	48.01	55.06	27.12	58.96	51.99	42.57	78.60	48.88	78.60	50.74
	LwF-MC	97.57	14.83	11.48	0.26	22.78	0.00	0.12	0.06	0.04	13.09	1.78	0.00	40.88	18.40	2.46	56.28	14.92	56.28	17.50
	ILT	98.34	40.12	36.63	26.51	46.64	34.39	19.35	49.41	71.50	40.54	55.17	27.29	51.78	43.54	42.88	67.84	45.61	67.84	47.00
	CIL	97.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	54.68	6.48	54.68	9.49
	MiB	98.50	41.26	44.48	**30.67**	50.61	39.30	17.11	**53.91**	74.67	47.58	55.54	33.00	53.57	48.35	**44.92**	**82.00**	48.90	**82.00**	50.97
	SDR	98.56	54.17	42.20	30.58	50.77	35.45	20.21	51.16	72.51	**49.62**	56.13	26.55	59.52	**53.60**	42.36	78.45	49.56	78.45	51.37
	Ours	**98.70**	**65.42**	**46.20**	30.29	**53.99**	**41.83**	**27.98**	53.29	**75.44**	38.93	**57.73**	**33.38**	**65.06**	51.58	44.56	77.43	**52.29**	77.43	**53.86**
Disjoint	SDR	**97.72**	66.25	**47.26**	1.53	44.14	32.53	0.59	47.95	18.41	20.12	52.21	26.78	60.41	51.77	**44.54**	**14.03**	40.81	**14.03**	39.14
	Ours	96.98	**68.74**	43.37	**6.56**	**48.93**	**38.17**	**4.67**	**48.40**	**40.15**	**31.40**	**57.22**	**32.90**	**60.45**	**51.79**	41.45	9.26	**44.75**	9.26	**42.53**
Overlap	SDR	97.86	63.42	**40.47**	3.52	47.69	40.46	0.21	49.63	33.52	29.77	53.35	36.05	**67.79**	51.70	44.60	18.14	44.00	18.14	42.39
	Ours	**98.12**	**71.15**	37.60	**10.12**	**53.88**	**42.48**	**6.97**	**51.35**	**60.66**	**34.30**	**56.04**	**37.58**	56.78	**61.49**	**44.86**	**23.19**	**48.23**	**23.19**	**46.67**
No-CIL	offline	98.85	75.00	54.34	31.59	57.14	46.43	23.54	54.50	78.40	44.87	61.34	39.27	68.94	60.27	46.64	86.37	56.07	86.37	57.97

* Abbreviation of ISAID - BG: background; BD: baseball diamond; BC: baseball court; BR: bridge; GTF: ground track field; HA: harbor; HC: helicopter; LV: large vehicle; PL: plane; RD: roundabout; SH: ship; SV: small vehicle; SBF: soccer ball field; ST: storage tank; SP: swimming pool; TC: tennis court.

TABLE X

PERFORMANCE COMPARISONS ON THE ISAID DATASET FOR 5-5-5 (THREE TASKS) CLASS-INCREMENTAL LEARNING SCENARIOS. BEST IN BOLD. OFFLINE IS NOT AN INCREMENTAL LEARNING METHOD, WHICH IS GENERALLY CONSIDERED TO BE AN UPPER BOUND ON PERFORMANCE IN CLASS-INCREMENTAL LEARNING

Protocols	Method	IoU Per Category(%)																mIoU (%)		
		BG	BD	BC	BR	GTF	HA	HC	LV	PL	RD	SH	SV	SFB	ST	SP	TC	old	new	all
Overlap	FT	97.61	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**33.22**	**50.82**	**62.25**	36.53	**83.41**	8.87	**53.25**	22.74
	LwF	97.18	51.21	30.56	28.10	32.72	34.08	0.22	41.81	66.41	20.42	53.32	30.55	37.47	56.89	**39.5**	81.12	41.46	49.11	43.85
	LwF-MC	97.79	51.63	24.44	6.06	12.93	0.58	0.00	9.20	30.15	**24.08**	12.80	5.54	49.14	42.02	15.31	65.80	24.51	35.56	27.97
	ILT	97.80	55.56	25.27	29.21	26.20	32.78	0.00	34.72	60.16	20.27	47.24	22.32	36.10	50.72	34.74	72.03	39.02	43.18	40.32
	CIL	97.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.46	34.32	40.96	26.74	71.47	8.83	41.99	19.19
	MiB	97.60	34.16	29.78	27.30	23.16	**46.20**	0.00	47.25	**72.74**	12.59	55.16	31.01	32.1	60.48	38.92	82.60	40.54	49.02	43.19
	SDR	98.13	50.01	**35.82**	**30.94**	34.36	36.66	0.03	44.80	70.09	18.43	52.56	31.12	41.68	56.66	37.10	81.13	42.89	49.54	44.97
	Ours	**98.27**	**59.63**	33.90	29.24	**36.07**	42.58	**0.31**	**47.36**	70.92	15.30	**55.54**	31.27	45.43	57.81	36.89	81.62	**44.47**	50.60	**46.38**
disjoint	SDR	97.29	44.30	**37.46**	30.51	**15.54**	17.59	1.87	40.70	66.61	**2.10**	37.29	30.69	21.15	**56.24**	**36.63**	**77.05**	35.57	44.35	38.31
	Ours	**97.51**	**45.53**	36.58	**32.15**	12.81	**23.19**	**2.93**	**42.75**	**68.24**	0.15	**44.69**	**31.07**	**28.47**	55.68	35.02	76.79	**36.96**	**45.41**	**39.60**
No-CIL	offline	98.85	75.00	54.34	31.59	57.14	46.43	23.54	54.50	78.40	44.87	61.34	39.27	68.94	60.27	46.64	86.37	56.07	86.37	57.97

the old task, outperforming the baseline method by more than 2% mIoU. In terms of performance on new tasks, MiB has the best performance, followed by LWF. Incremental learning aims to measure the performance of the model on all classes. Our method leads the baseline method by 2.49% mIoU. The next best method is the baseline method. Our method has the highest IoU on almost all categories. Therefore, in the disjoint and overlap scenarios, we only experiment with the baseline method and our method, and the IoU for each category is given in Table IX. Our method outperforms the baseline method by 4% mIoU, which proves the effectiveness of our method.

In the third experimental setting, the model is optimized three times incrementally: five categories are learned for the first time and five categories are learned each time for the latter two tasks. The experimental results are shown in Table X. We conduct experiments on disjoint and overlap protocols. Our method outperforms the baseline method SDR 1.58% and 1.41% mIoU on the old task and overall, respectively. Our method is inferior to FT in performance on new tasks but outperforms other compared methods. The reason for this may be that FT, as a transfer learning method, completely forgets the knowledge learned by the old tasks and classifies almost all instances into the foreground category of the new

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TABLE XI

PERFORMANCE COMPARISONS ON THE GAOFEN CHALLENGE DATASET FOR 5-3 TASKS. BEST IN BOLD. OFFLINE IS NOT AN INCREMENTAL LEARNING METHOD, WHICH IS GENERALLY CONSIDERED TO BE AN UPPER BOUND ON PERFORMANCE IN CLASS-INCREMENTAL LEARNING

Protocols	Method	IoU Per Category(%)									mIoU(%)		
		BG	water	road	building	vehicle	land	ground	tree	lawn	old	new	all
mib	FT	0.16	0.00	0.00	0.00	0.00	0.00	**63.65**	**89.81**	72.67	0.03	**75.38**	25.14
	LwF	0.37	79.05	70.37	80.23	48.91	41.63	61.18	89.56	70.24	53.43	73.66	60.17
	LwF-MC	0.22	62.17	47.27	63.93	31.89	24.85	40.69	81.80	56.91	38.39	59.80	45.53
	ILT	**0.63**	80.93	70.34	79.96	46.72	42.53	61.75	89.05	71.05	53.52	73.95	60.33
	CIL	0.07	0.00	0.00	0.00	0.00	0.00	63.11	89.81	72.43	25.05	75.12	25.05
	MiB	0.00	74.89	70.17	81.42	53.74	43.18	62.62	89.33	71.82	53.90	74.59	60.80
	SDR	0.00	72.09	70.48	**82.08**	**55.67**	**44.44**	61.49	89.53	69.78	54.13	73.60	60.62
	Ours	0.48	**82.38**	**71.06**	81.86	50.36	44.18	63.31	89.62	**72.68**	**55.05**	75.20	**61.77**
overlap	FT	0.20	0.00	0.00	0.00	0.00	0.00	61.8	**88.46**	**71.44**	0.03	**73.90**	24.66
	LwF	1.21	**83.76**	70.56	82.39	46.44	45.00	62.08	88.12	70.9	54.89	73.70	61.16
	LwF-MC	0.24	65.42	33.84	47.98	19.06	18.33	45.97	78.28	54.78	30.81	59.68	40.43
	ILT	1.02	81.96	70.24	81.55	44.97	46.23	61.19	87.43	70.19	54.33	72.94	60.53
	CIL	0.13	0.00	0.00	0.00	0.00	0.00	61.89	88.19	70.9	24.57	73.66	24.57
	MiB	0.00	82.17	71.47	83.25	51.73	**48.76**	62.39	88.27	70.96	56.23	73.87	62.11
	SDR	0.00	82.81	72.01	83.39	**51.99**	48.49	**62.45**	88.12	70.54	56.45	73.70	62.20
	Ours	**1.43**	83.57	**72.89**	**83.52**	51.35	48.56	62.17	88.04	71.02	**56.89**	73.74	**62.51**
No-CIL	offline	15.96	82.85	71.02	82.60	53.97	50.13	62.58	88.95	72.33	59.42	74.62	64.49

* Abbreviation of GaoFen challange data set - BG: background; ground: Impervious ground

task. The baseline method is only worse than our method in old task performance, new task performance, and overall performance of 5-5-5 tasks on the iSAID dataset. In the disjoint protocol, we only conduct experiments on the baseline method and our method. Our method outperforms SDR by 1.39% mIoU, 1.06% mIoU, and 1.29% mIoU in performance on old tasks, performance on new tasks, and overall performance, respectively. Remarkable experimental results demonstrate the effectiveness of our method. Retaining similar and important information between the new model and the old model through the prototype helps the model to incrementally learn more knowledge of semantic categories. Incremental learning solves the problem that the model learns more knowledge only by applying new and old data to the model for more resource-intensive training.

2) Experiments on the GaoFen Challenge Dataset: We conduct experiments on the GaoFen challenge dataset to demonstrate the effectiveness of our method. Similar to ADE20K, the pixels of the GAOFEN challenge dataset whose label is assigned as background are small. There are only a dozen images in some tasks in the disjoint protocol. Therefore, we conduct experiments on the 5-3 and 7-1 task settings in the overlap and mib protocols. The 5-3 task setting means that the model is trained to learn to segment five foreground categories in the first task, and the second task learns three foreground categories. The experimental results are shown in Tables XI and XII. Our methods have achieved the best results under mib and overlap protocols, especially mib.

The categories contained in the Gaofen Challenge dataset are less intraclass dissimilar, and the scale and morphology of the same semantic category vary little. Coupled with the overlap setting, the data of the current task contain all images of the current semantic category. Therefore, it is more accurate to calculate the prototype of the current category in the new task. Considering that the change of the model parameters will still affect the value of the feature vector, the true class center will have a slight shift. Therefore, our proposed method still improves the overlap protocol for datasets with few pixels belonging to the background class. When the semantic categories contained in each task are not all, the difference of objects of different colors and shapes still affects the prototype. Furthermore, the numerical fluctuations due to model updates cannot be underestimated. Therefore, under the mib experimental setting, no matter which dataset is used, excellent experimental results are achieved.

3) Experiments on the Pascal VOC2012 Dataset: The experimental results of different methods under the 10-5-5 setup are shown in Fig. 8. The overall performance of

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TABLE XII

PERFORMANCE COMPARISONS ON THE GAOFEN CHALLENGE DATASET FOR 7-1 TASKS. BEST IN BOLD. OFFLINE IS NOT AN INCREMENTAL LEARNING METHOD, WHICH IS GENERALLY CONSIDERED TO BE AN UPPER BOUND ON PERFORMANCE IN CLASS-INCREMENTAL LEARNING

Protocols	Method	IoU Per Category(%)									mIoU(%)		
		BG	water	road	building	vehicle	land	ground	tree	lawn	old	new	all
mib	FT	0.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	74.87	0.02	74.87	8.34
	LwF	0.82	65.25	70.58	81.55	52.63	40.26	60.26	82.28	75.37	56.70	75.37	58.78
	LwF-MC	0.25	17.18	53.03	64.03	35.4	20.23	48.79	57.01	57.79	36.99	57.79	39.30
	ILT	**1.36**	63.48	70.33	81.67	51.2	39.87	59.64	83.37	75.31	56.37	75.31	58.47
	CIL	0.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	75.14	0.02	75.14	8.37
	MiB	0.00	62.43	69.77	81.48	52.60	**43.11**	60.06	82.54	74.96	56.50	74.96	58.55
	SDR	0.00	62.08	70.44	81.68	**52.78**	42.66	60.82	**83.6**	73.60	56.76	73.60	58.63
	Ours	1.19	**67.08**	**71.52**	**82.48**	52.47	42.18	**60.65**	82.38	**75.46**	**57.49**	**75.46**	**59.49**
overlap	FT	0.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	71.97	0.01	71.97	8.01
	LwF	1.82	83.44	71.09	82.29	51.91	44.02	61.39	88.59	71.78	60.57	71.78	61.81
	LwF-MC	0.20	40.5	53.53	69.42	32.74	16.5	48.66	44.54	56.82	38.26	56.82	40.32
	ILT	1.58	82.47	71.57	82.33	50.96	44.06	61.85	88.69	71.86	60.44	71.86	61.71
	CIL	0.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**72.28**	0.01	**72.28**	8.04
	MiB	0.00	83.06	71.35	81.86	52.36	45.53	61.33	89.03	72.20	60.57	72.20	61.86
	SDR	0.00	**84.33**	**71.85**	82.32	51.56	**47.99**	61.77	**89.07**	72.21	61.11	72.21	62.34
	Ours	**6.52**	84.09	71.2	**82.71**	**52.99**	44.49	**62.15**	88.37	71.45	**61.57**	71.45	**62.66**
No-CIL	offline	15.96	82.85	71.02	82.6	53.97	50.13	62.58	88.95	72.33	59.42	74.62	64.49

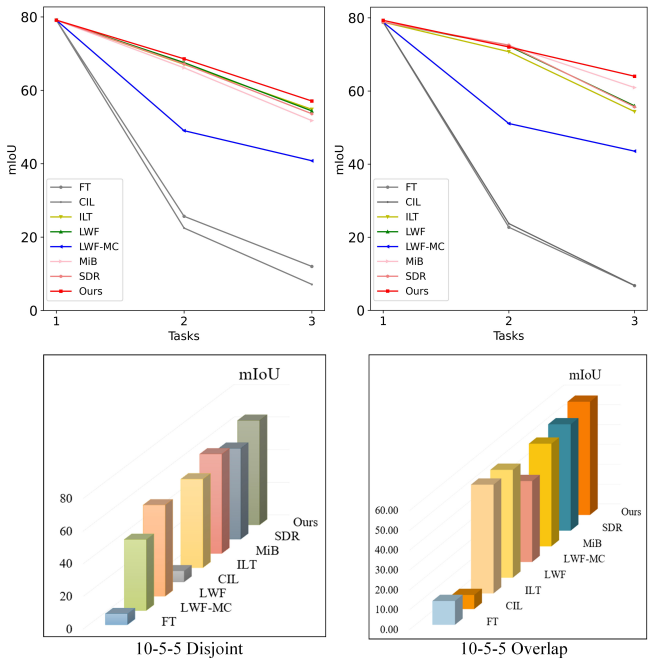


Fig. 8. Visualization results of Pascal VOC2012.

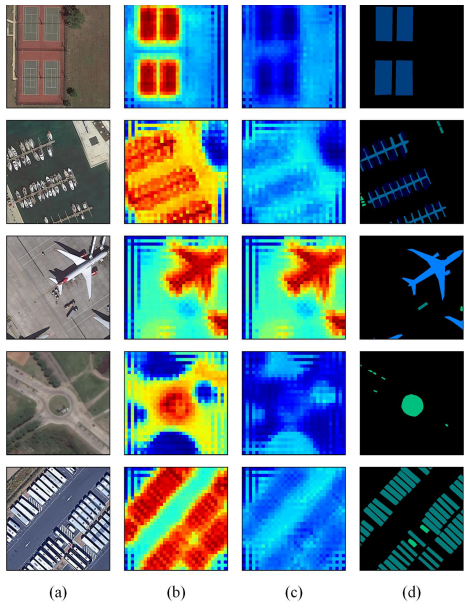


Fig. 9. Visualization results of similarity between the prototype and feature vectors of the same and different categories. (a) Input image. (b) Similarity between prototype and corresponding feature vector. (c) Similarity between the prototype of plane and the feature vector. (d) Ground truth.

FT and CIL is the worst among all methods. LWF-MC introduces the single-head knowledge distillation method, and the effect is improved. LWF is a multihead knowledge distillation method, which is better than LWF-MC. The experimental performance of ILT, MiB, and SDR has different degrees of improvement for LWF. Our method exceeds SDR by 3.46%

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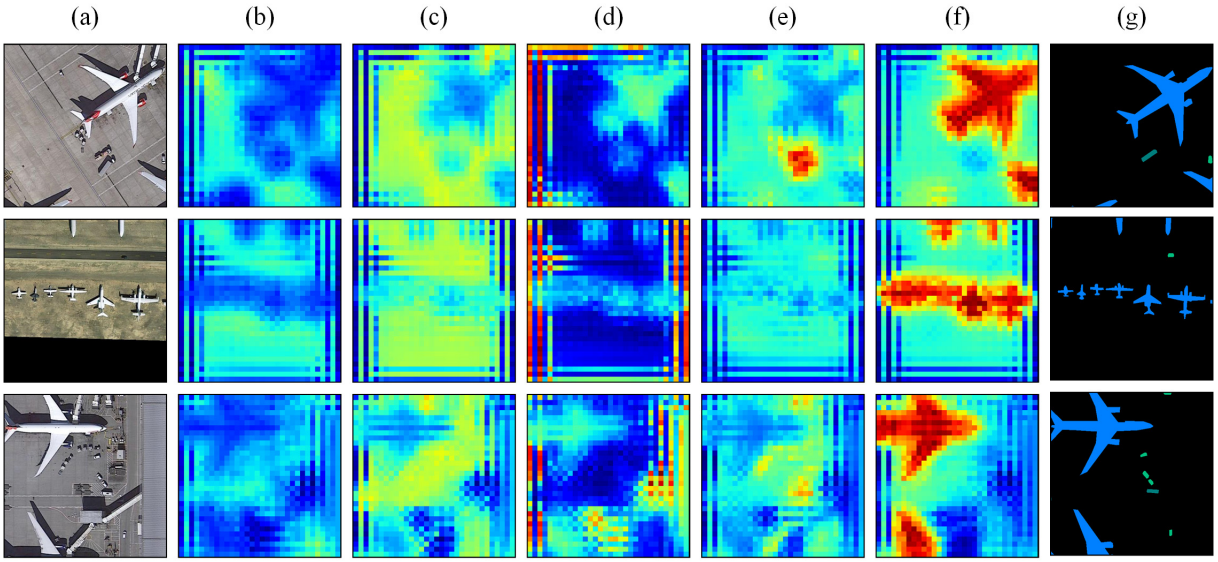


Fig. 10. Visualization results of similarity map between feature vectors and prototypes of the same and different categories. (a) Input image. (b) Similarity between prototype of tennis court and feature vectors. (c) Similarity between prototype of harbor and feature vectors. (d) Similarity between prototype of roundabout and feature vectors. (e) Similarity between prototype of large vehicle and feature vectors. (f) Similarity between prototype of plane and feature vectors. (g) Ground truth.

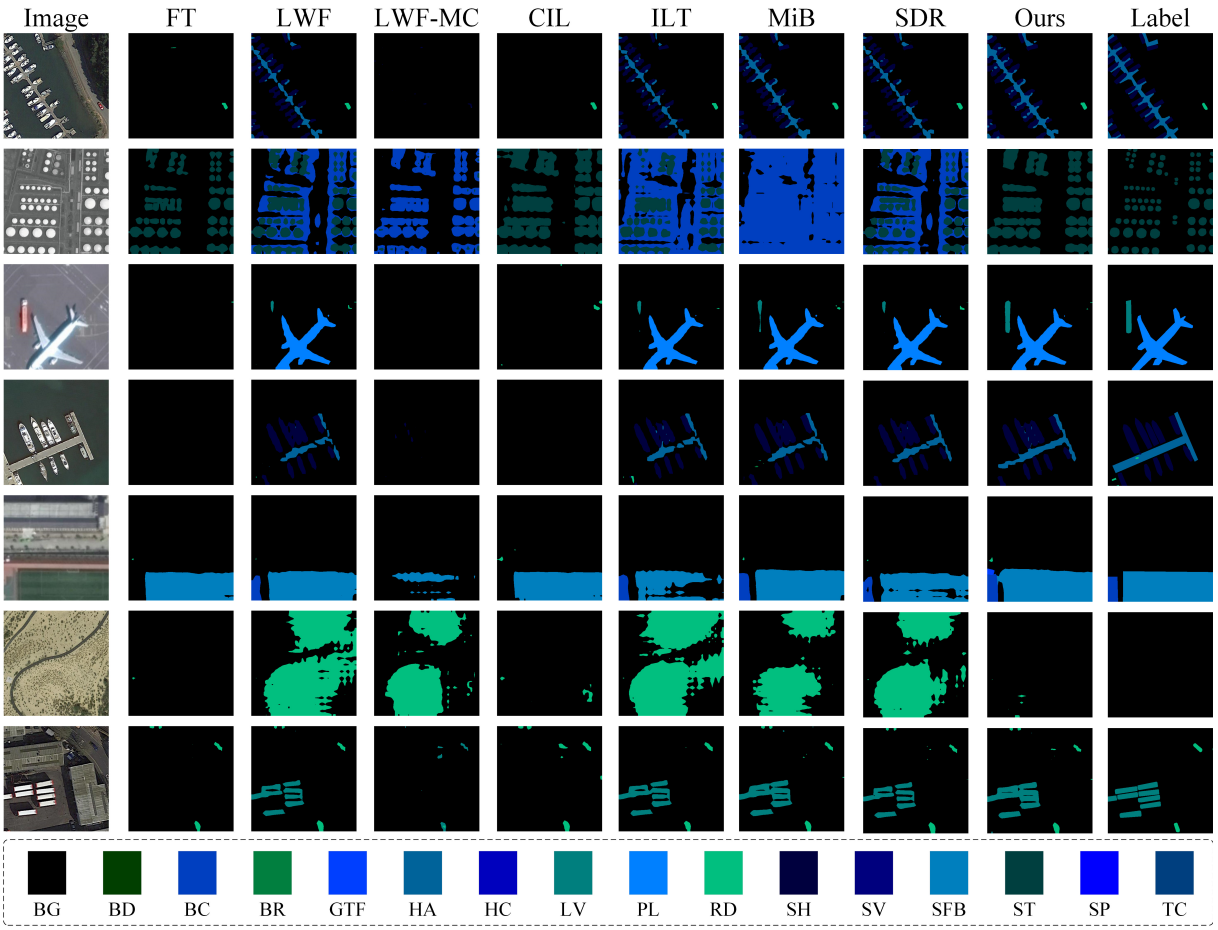


Fig. 11. Qualitative comparisons between our method and compared methods on the iSAID dataset.

mIoU and 8.38% mIoU in the disjoint and overlap protocol, respectively. Our method has distinguished performance compared to the baseline method, which proves the effectiveness of our method.

F. Qualitative Analysis

1) Experiments on the iSAID Dataset: As further qualitative analysis of feature vectors and prototypes, we visualize the similarity between the prototype and feature vectors of the

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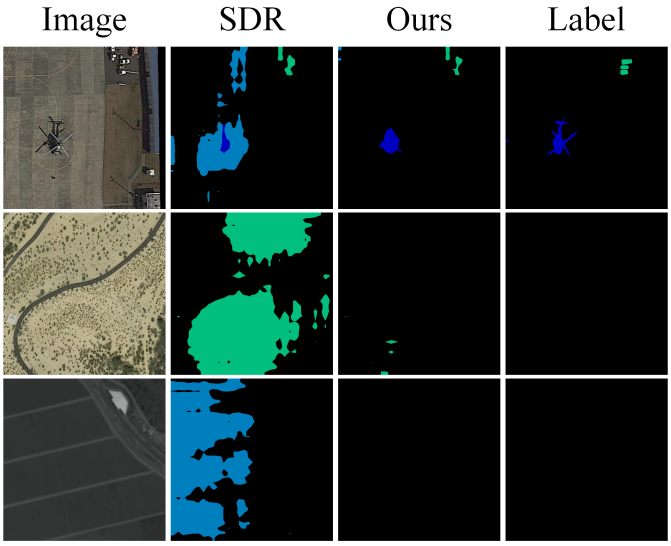


Fig. 12. Visualization results of false alarms on the iSAID dataset. Our method alleviates the false alarm problem.

same and different categories, which is shown in Fig. 9. The redder the content in the image, the greater the similarity. Fig. 9 proved that the prototype of our method has a greater similarity with the feature vector of its own category, and the similarity with the feature vector of other categories is very small, even smaller than the similarity with the background class. This also shows that the prototype can represent the global information of semantic categories and is an effective medium for preserving old knowledge between new models and old models.

Fig. 10 shows the similarity of feature vectors for different classes of prototypes and different shapes of aircraft classes. Objects such as large vehicles are also included in each image. In Fig. 10(e) of the first row, the feature vector of the corresponding location and the prototype of the large vehicle also have the largest similarity (red). Fig. 10 proves that the planes corresponding to different shapes all satisfy the large intraclass similarity between the prototype and the feature vector, and the small interclass similarity. In remote sensing scenarios, it is reasonable to use a prototype as a representative feature vector to retain the same representative information between the new model and the old model so that the model does not forget the old knowledge while learning new knowledge.

The qualitative results on the iSAID dataset are shown in Fig. 11. FT performs poorly on the old task. The model predicts almost all the positions of the old task as background, but it performs well on the new task because it does not pay attention to the performance of the old task so that the model can be trained using the new data as much as possible, as shown in the second and fifth rows. LWF-MC and CIL also perform poorly on the old task. LWF and ILT have remarkable performance in alleviating the catastrophic forgetting problem, but they have wrong predictions for pictures that are all backgrounds. MiB and SDR also suffer from false alarm problems, but their overall performance is better than that of LWF and ILT. Our method has prominent performance in

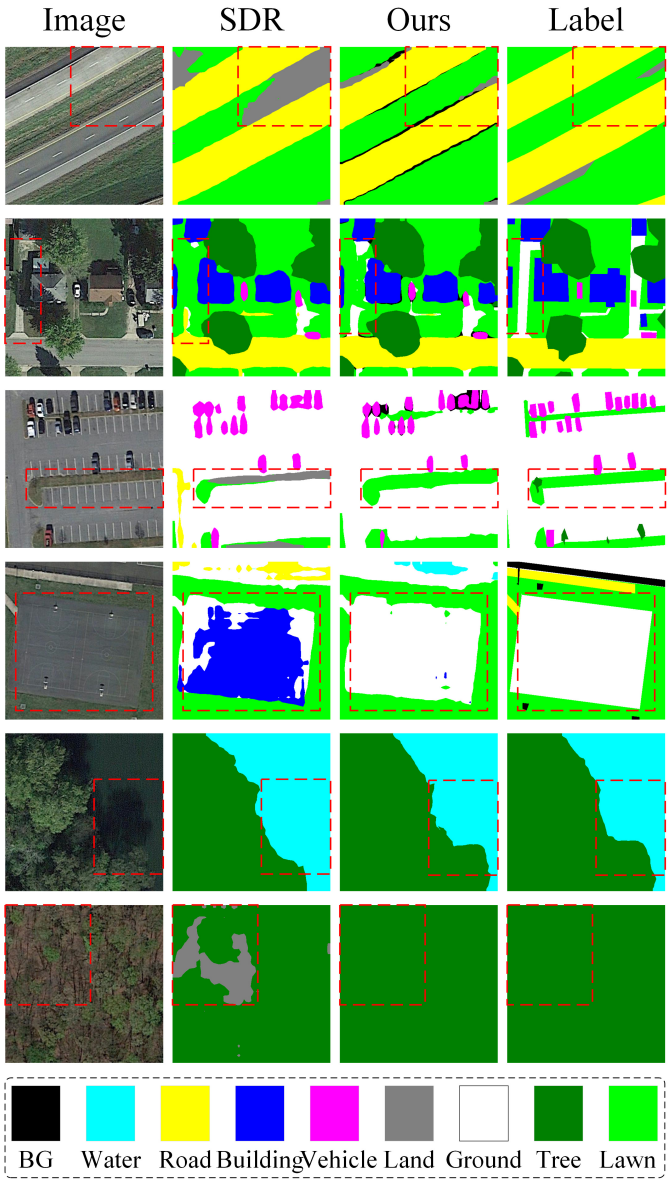


Fig. 13. Qualitative comparisons between our method and baseline method on the Gaofen Challenge dataset.

mitigating the false alarm problem, as shown in Fig. 12. Our method performs well when the labels contain a small number of pixels that need to be classified or none at all. We consider the appearance of this phenomenon to be related to the fact that we decrease the similarity of prototypes and feature vectors of different classes. Overall, the visualization results of our method are the closest to the labels, which is also consistent with the quantitative results. Remarkable qualitative analysis results demonstrate the effectiveness of our method.

2) Experiments on GaoFen Challenge Dataset: The qualitative results of the Gaofen Challenge dataset are shown in Fig. 13. In many categories, our approach is better, such as water and impervious ground. The more excellent ways are used to emphasize with a red dotted line. The superior parts of our method are highlighted with red dashed boxes.

3) Experiments on Pascal VOC2012 Dataset: We further conduct experiments on the Pascal VOC2012 dataset, and

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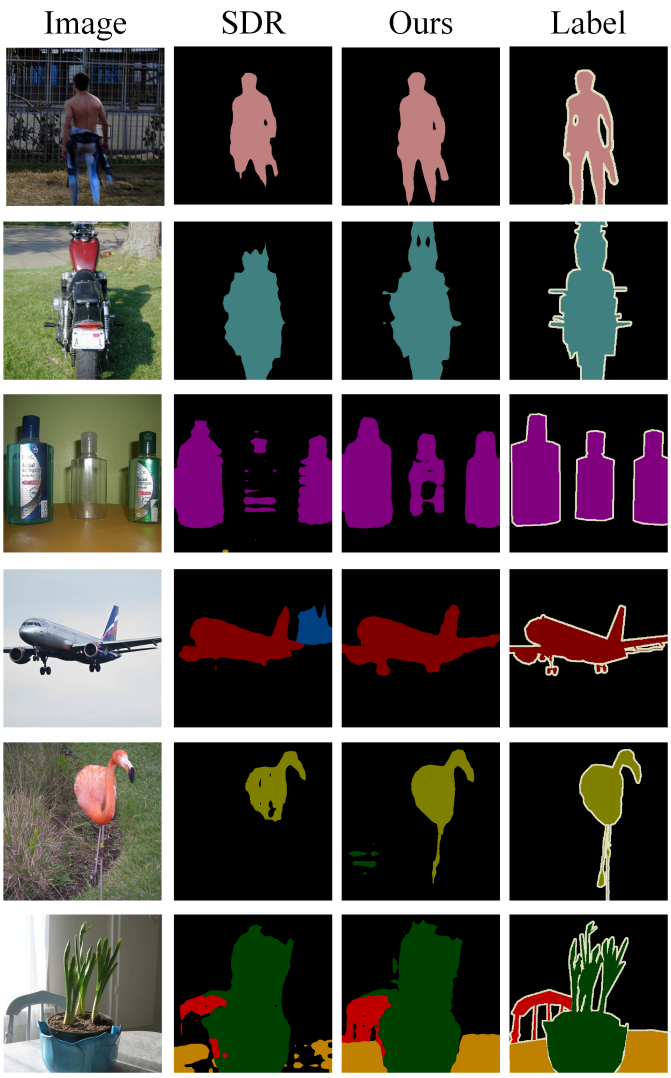


Fig. 14. Qualitative comparisons between our method and baseline method on the Pascal VOC2012 dataset.

the experimental results are shown in Fig. 14. Our method performs well in multiple categories. For the person class, we can more accurately distinguish human legs. For the plane category, our method also has good segmentation results on pulleys, and SDR also has a small fraction of misclassifications around planes. For tables, we distinguish more accurately than SDR. Experiments on a more general semantic segmentation dataset further demonstrate the effectiveness of our method.

V. Conclusion

In this article, we propose a novel SIL method based on the impact of inconsistent LAND on the problem of catastrophic forgetting. It is observed that it is important to bridge the feature gap between incremental learning methods and static learning in remote sensing. We use prototypes as a medium for preserving knowledge of old tasks between old and new models. We propose a prototype update module to make the number distribution of high-level semantic labels involved in prototype computation close to that of static learning. Moreover, we propose the SMM, which constrains the semantic consistency of feature vectors and the corresponding prototypes to achieve greater intraclass similarity. Finally, we propose the LR to preserve interclass context relation. Experiment results on multiple datasets demonstrate the effectiveness of our approach. In the future, our research will focus on exploring why different orderings of categories lead to different final results. We will propose an evaluation metric that measures the impact of different orders on the final result. Besides, multiple prototypes for each category will be used to ensure the continuation of old knowledge.

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